Palaeomagnetic Data from South America

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1. Introduction

Two field trips of duration about 3 months were made in 1956 and 1958 and a collection of rock samples spanning the geological column from sites in Brazil, Argentina, Uruguay, Venezuela, Colombia, Peru and Chile was made. Help from local Geological Surveys, Universities, from the Shell, Petrobras and Y.P.F. Oil Companies, from mining companies and from many Government agencies is gratefully acknowledged.

A brief account of the preliminary results of these surveys is given here, and details will be published elsewhere later. The work on the Quaternary basaltic flows of W. Argentina and on the Serra Geral Formation of Triassic-Jurassic age has already been reported by Creer (1958) and a fuller account of the results from the basaltic flows and diabase dikes of the Serra Geral Formation is now in the press by Creer (1962).

Neither of the above results reveals appreciable polar movement assuming an axial dipolar geomagnetic field in the past. Both of these formations are igneous and the natural remanent magnetization is of thermal origin so there are no doubts concerning the time of magnetization. Furthermore after cleaning in alternating magnetic fields, both normal and reversed directions separated by very nearly 180° are found and this indicates stability of magnetization.

However there is strong geological evidence of glaciation in the Permo-Carboniferous, particularly in Southern Brazil. Similar evidence of glacial conditions in these times exists in Africa and Australia and for this, among many other reasons, several geologists, of whom Wegener and Du Toit are perhaps the most famous, have suggested the existence of the gigantic primeval continent Gondwanaland before the commencement of the Mesozoic. It is probable that the remanent magnetism of rocks might reveal independent evidence relevant to this hypothesis and this was one of the objects of the two palaeomagnetic surveys undertaken.

At present fairly good palaeomagnetic data has been published by Irving and coworkers for the continent of Australia. However, because of their relatively larger areas, it is of fundamental importance that reliable palaeomagnetic data be obtained from rock formations in both Africa and S. America.

The productivity of the palaeomagnetic surveys of both these large continents (by Nairn in Africa) has been poor compared with that for Europe and North America. Under the oxidizing conditions which frequently exist in tropical environments, magnetization of new magnetic minerals produced by weathering processes or recrystallisation

POLE POSITIONS AND DIRECTIONS OF MAGNETIZATION FOR ROCK FORMATIONS FROM S. AMERICA

(discs assigned unit weight except where items are bracketed when samples have been given unit weight)

Age and Locality	Pole Position		Radii of 95% Confidence			Directions of Magnetization		Radius of 95% confidence	Number of discs or samples	Mean Site Position	
	Long.	Lat.	Circle	Oval		D	I			Long.	Lat.
1. Quaternary Lavas	148°.3W 119° E	69°.1N 82°.3N	— —	(11°.8) (27°.4)	(7°.9) (21°.8)	350° 188°	−46° +64°	7°.8 5°.2	12 46	71°W	38° S
2. Tertiary Lavas	107°.5E	69°.4N	—	(6°.8)	(6°.0)	+181°.0	+73°.0	3°.8	—	71°W	38° S
3. Huitrinian (Neuquen, Argentina)	(118°.3W)	(65°.5N)	(13°.5)			(340°.9)	(−37°.8)*	(13°.5)	(13)	71°W	39° S
4. Kimmeridgian (Neuquen, Argentina)	(7°.6W)	(86°.4N)	(22°.7)			(4°.1)	(−57°.2)*	(22°.7)	(7)	71°W	39° S
5. Serra Geral (N+R)	125°.9W	78°.2N	(5°.7)			(347°.2)	(−37°.6)	(5°.7)	(30)	50°W	25° S
6. Las Cabras (Mendoza, Argentina)	(63°.6W)	(71°.1N)	(26°)			(1°.7)	(−26°.7)*	(26°.4)	(11)	69°W	33° S
7. La Quinta (Triassic?)	3°.3W	17°.5N	—	(29.°4)	(17°.9)	(+254°.3)	(−41°.1)*	(24°.2)	(18)	71°W	8° N
7b. Giron (Triassic ?)	49°.9W	82°.1N	—	(17°.9)	(9°.7)	(3°.4)	(+26°)*	(16°.6)	(27)	74°.5W	6°.5N
8. Triassic (La Rioja, Argentina)	26°.2E	46°.6N	—	22°.3 (51°.6)	12°.7 (29°.1)	46°.2 (41°.7)	−33°.0 (−32°.6)	19°.7 (45°.6)	13 (4)	65°W	24° S
	65°.3E	71°.8N	—	12°.2 (23°.5)	8°.6 (16°.9)	196°.9 (197°.9)	(+54°.3 +55°.9)	8°.7 (16°.4)	49 13		
9. Corumbatai (Permian) (Parana, Brazil)	((38°.3W))	((82°.2N))	—			((180°))	((+60°))*	—	—	50°W	25° S
10. Mitu Formation (Upper Carboniferous or Permian) (Peru)	43°.5E	19°.7N	13°.7	14°.9	9°.3	76°.1 (76°.2)	−43°.1 (−44°.0)	11°.9 (17°.1)	23 (8)	75°W	12° S
	56°.8E	18°.3N	16°.8	20°.0	14°.8	258°.0 (257°.6)	+58°.4 (+59°.6)	13°.5 (23°.2)	57 (19)		
11a. Mississippian (Peru)	((85° E))	((8° S))	—	—	—	((315°))	((+75°))	—	12	75°W	12° S

continued

11.b Mississippian (La Rioja)	115° E	60° N	—	—	—	180°	+55°	—	9	65°W	24° S
12. Devonian (Uruguay)	(131°.5E)	(76°.7S)	((11°))	—	—	(1°.7)	(−35°.7)*	(34°.2)	(6)	55°.5W	30° S
13. Devonian (Salta)	142°.8E	31°.2S	(20°.8)	12°.4 (23°.1)	7°.9 (15°.0)	26°.3 (27°.5)	+46°.1 (+47°.4)	9°.7 17°.8	75 (22)	65°.5W	24° S
14. Devonian-Silurian (Parana, Brazil)	(36°.2E)	(62°.1S)	—	—	—	149°.5	+40°.7*	—	(3)	50°W	25° S
15. Jacadigo (Silurian) (Corrego das Pedras) (Corumba, Brazil)	179°.8E	18°.7S	—	6°.7	4°.0	59°.8	+38°.8	19°.4	13	58°W	19° S
16. Cambro-Ordovician (Salta, Argentina)	144°.7E	51°.9S		15°.0 (26°.7)	7°.8 (13°.8)	18°.3 (23°.5)	+17°.8 (+17°.0)	14°.5 (25°.8)	76 (20)	65°.5W	24°.5S
17. Cambrian (Salta, Argentina)	(152°.1E)	(48°.9S)		— (49°.2)	— (25°.6)	(24°.0)	(+18°.2)	(47°.4)	— (17)	65°W	24.5° S
18. Pre-Cambrian (Jujuy, Argentina)	133°.9E	44°.3S	15°.7	17°.1 (27°.7)	9°.6 (15°.6)	193°.6 195°.9	−31°.9 −32°.2	15°.2 (24°.6)	21 (7)	65°W	25° S
19. Pre-Cambrian, (Guabiroba, Parana, Brazil)	147°.8E	6°.3S	—	—	—	209°.9	−70°.5	—	(3)	49°W	25° S

Footnote to Table

1. Disks have been assigned unit weight except where entries are included in brackets when the mean directions of hand samples have been assigned unit weight in computing mean directions of magnetization of formations. Entries included in double brackets are estimated directions of primary magnetization of formations in which the directions of magnetization of disks show pronounced streaking.
2. ※; these formations are likely to have been remagnetized in recent geological time.
3. N.B. In all the stereograms, the points represent the directions of the north seeking poles of remanent magnetization. Directions plotted thus, ○, are from upper hemisphere and thus, ●, from lower hemisphere. The primitive always represents the bedding planes. Mean directions marked thus ⊗ and ×. Direction of present field ◇ and of present axial dipole field □ and ■.

magnetization of pre-existing minerals in the direction of the contemporary geomagnetic field destroys all apparent evidence concerning the direction of the palaeo-geomagnetic field. Methods by which the destruction of such recent magnetization can be undertaken without appreciably affecting the fossil magnetization have not yet been perfected for sedimentary rocks.

The precision of the data given in the following sections might be improved by magnetic cleaning, but the problem of extending these techniques to weakly magnetized rocks is not yet solved. However, secondary magnetization often reveals itself by the presence of streaking (e.g. see Creer, 1957) and where both normally and reversely magnetized sites occur in a given formation the effects of secondary magnetization can usually be satisfactorily allowed for.

Directions of magnetization, circles and ovals of confidence at the 95 % level assuming Fisherian distributions and pole positions assuming dipole fields are given in the table, together with formation names, site positions and geological period. The results are considered, first for Mesozoic and younger formations in Section 2 and then for Palaeozoic and Precambrian formations in Section 3.

2. Magnetization of Mesozoic, Tertiary and Quaternary Formations

The most recent formations studied are the Quaternary and Tertiary basaltic flows from Neuquen Province in W. Argentina. The Quaternary lavas are flat lying and samples were collected from several sites from between San Carlos de Bariloche in the south to Buta Ranquil in the north, just to the east of the Andean pre-cordillera. The Tertiary lavas were collected from four flows at one site in the north of this area and the flows are tilted at an angle of between 10° and 15° to the horizontal.

The Upper Cretaceous is represented by the purple sandstones of Huitrinian-age and the Lower Cretaceous by rocks of similar lithology and colour from the Kimmeridgian. The former exhibit only normal remanent magnetization, the mean direction being just significantly different from both the present geomagnetic field direction and from the direction of an axial dipole field (see fig. 1). The latter exhibit both normal and reversed magnetization, the normal direction being significantly different from the present geomagnetic field direction but close to the present axial dipole field, while the reversely magnetized samples exhibit streaking (see fig. 2).

The magnetization of the Las Cabras and Rio Blanco Formations from Mendoza Province, W. Argentina show appreciable scatter and the mean appears to lie closer to the present field direction than to the axial dipole field (see fig. 3). An inspection of this stereogram suggests that some of the directions shown might have had a reversed primary magnetization (some of the discs have downward inclinations of n.r.m.).

From Venezuela, a collection was made of the La Quinta formation which consists of red beds and is believed to be Triassic or possibly early Jurassic. The stereogram, fig. 4, shows considerable scatter of magnetization. This formation was appreci-

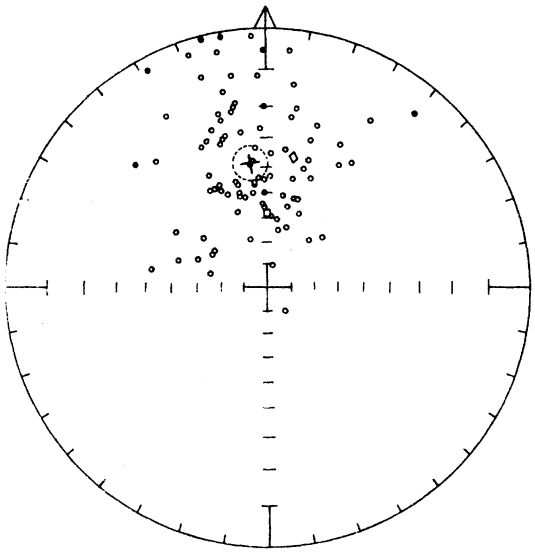


Fig. 1. Directions of n.r.m. of discs of purple Huitrinian sandstones (upper Cretaceous) from Neuquen province, Argentina.

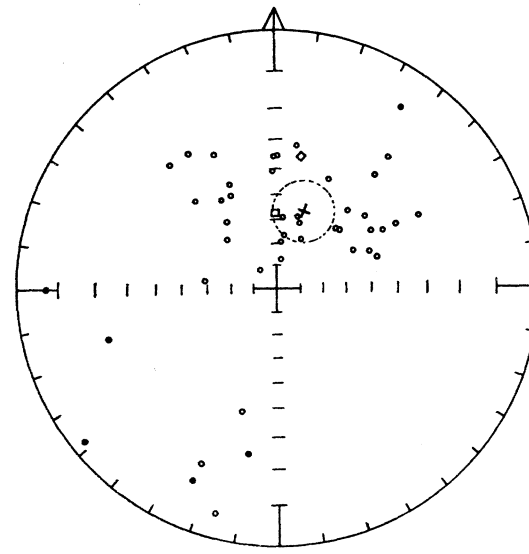


Fig. 2. Directions of n.r.m. of discs of purple Kimmeridgian sandstones (lower Cretaceous) from Neuquen province, Argentina.

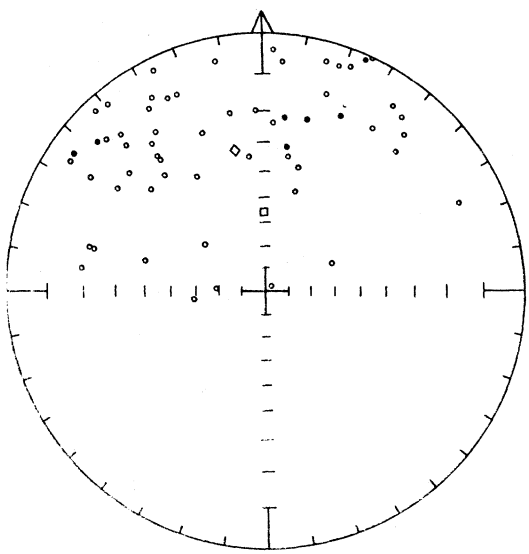


Fig. 3. Directions of n.r.m. of discs from the Las Cabras and Rio Blanco formations (probably Triassic) from Mendoza province, Argentina.

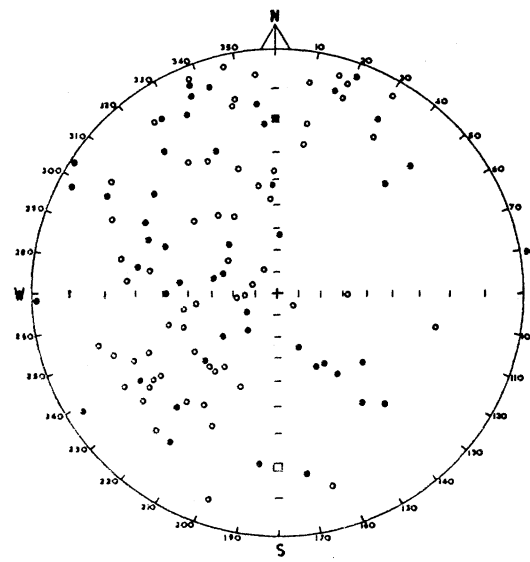


Fig. 4. Directions of n.r.m. of discs from the La Quinta formation (probably Triassic), Venezuela.

ably folded during the formation of the Venezuelan Andes. In fig. 5 are shown mean directions of magnetization of hand-samples (the directions of discs have been meaned) from the type locality at La Quinta where the geological tilt is small, of the order of 10°. The axial dipole field direction is shown and it is apparent that here also there is evidence of "streaking". The distribution of directions shown in this figure is not inconsistent with the suggestion that they were originally magnetized in a direction nearly opposite to the present axial dipole field.

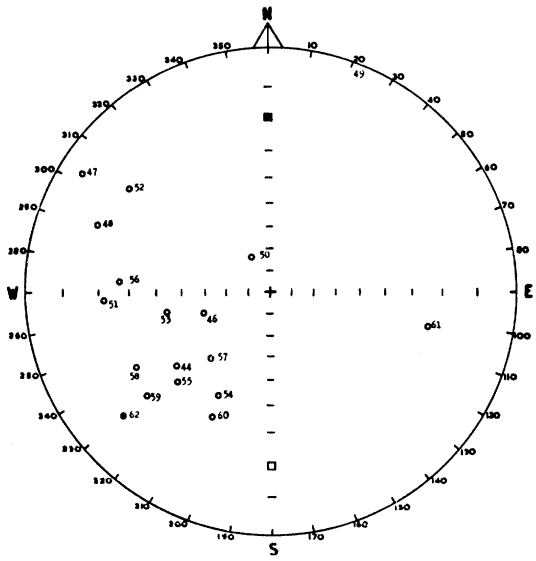


Fig. 5. Mean directions of n.r.m. of hand samples from the La Quinta type locality.

In Colombia, a formation of Mesozoic red beds equivalent to the La Quinta, is known as the Giron. Only normally magnetized samples have been observed and the mean direction is not significantly different from the present axial dipole field direction.

The magnetization of the Serra Geral Formation has been thoroughly investigated (Creer, 1958 and 1962). Normal and reversed directions are found in both lavas and dikes and the mean pole position given by the axis of magnetization is about 12° away from the geographic pole and just significantly different from it.

Triassic red beds exhibiting both normal and reversed magnetization have been collected from the provinces of Salta and Jujuy in N.W. Argentina and the results are shown in fig. 6. The mean pole positions for normal and reversed sets of discs are not exactly at 180°. Of these that for the reversed set of magnetizations is likely to be the more reliable since it is based on more measurements, but the best pole position will lie somewhere between the two given in the table and shown in the diagram of pole positions in fig. 13.

In Paraná State, Brazil, the red muds and clays of the Corumbatai Formation have been sampled. Below this formation lies grey clays, and below these glacial Carboniferous sandstones. Normal and reversed directions are found (see fig. 7), and the existence of the former is unusual since this formation is dated as Permian and no normally magnetized Permian rocks have been reported elsewhere. Both normal and reversed groups of data exhibit streaking. An estimated primary direction of magnetization has been made from the reversed group and is given in the table.

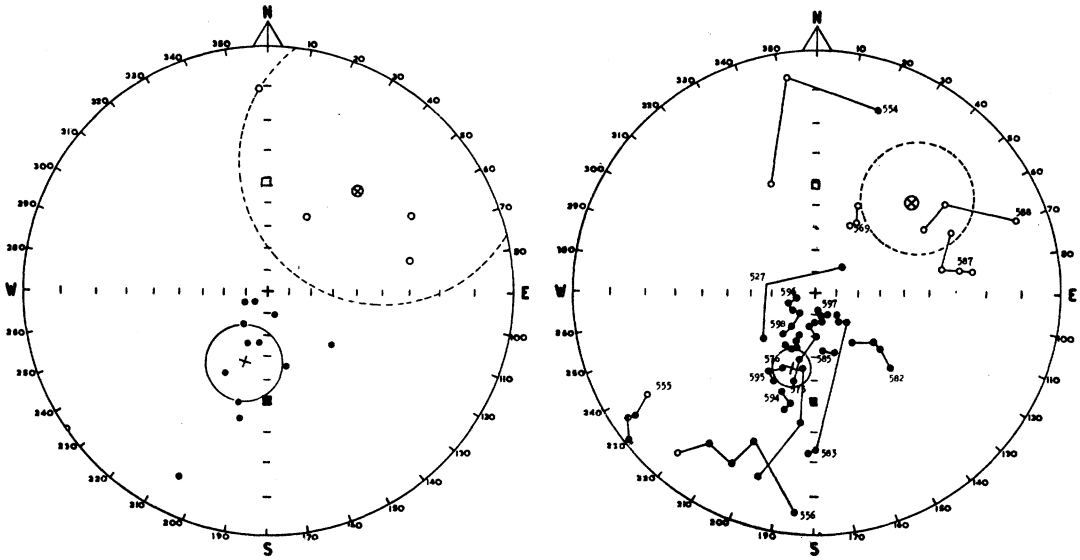


Fig. 6. Triassic red beds from Salta and La Rioja provinces, Argentina. Primitive of projection is bedding plane, ⊗, × mean directions of magnetization.
(a) Directions of magnetization of hand samples;
(b) Directions of magnetization of discs.

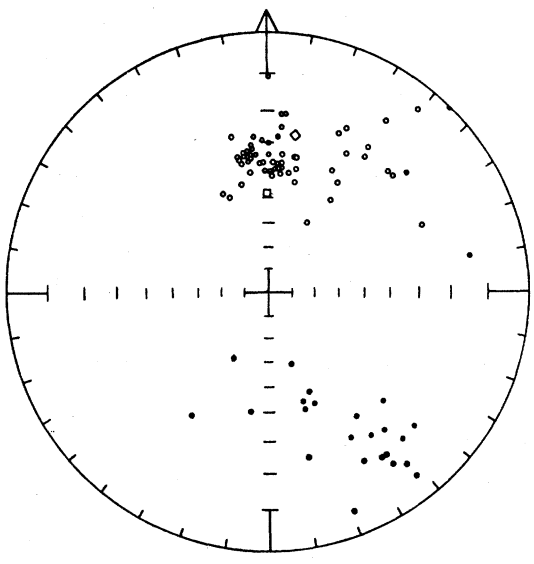


Fig. 7. Directions of n.r.m. of discs from the Corumbatai formation (upper Permian), from Sao Paulo and Parana states, Brazil.

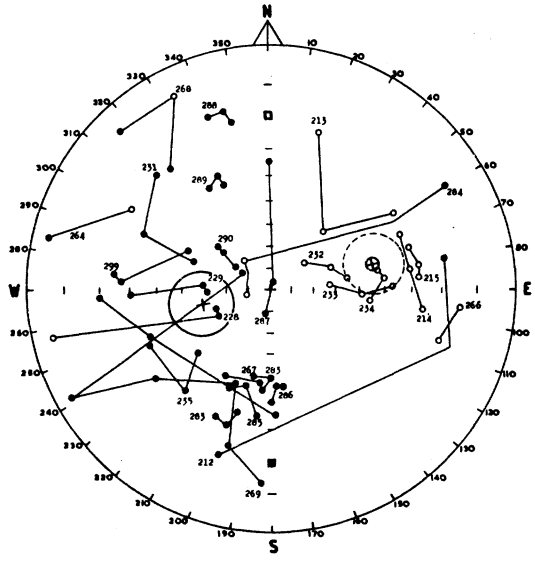


Fig. 8. Directions of discs from the Mitu formation (Permian), from Peru.

3. Magnetization of Palaeozoic and Precambrian Formations

The Mitu Formation of red beds in Peru is dated as Upper Carboniferous. Normal and reversed directions of magnetization are found (see fig. 8), the axis being significantly different from the present dipole field direction and normal and reversed magnetizations being nearly at 180°. The samples were collected from a number of sites in the W. Andes to the N and E of Lima and the geological tilt is appreciable. This suggests that the magnetization has been stable, at least since folding occurred and probably since the beginning of the Mesozoic for the evidence from section 2 suggests that the geomagnetic field-axis retained its present orientation with respect to S. America almost throughout the Mesozoic, only the Triassic of W. Argentina indicating polar movement. Moreover if the magnetization of the Mitu Formation (which is post Pennsylvanian) had been affected by this field the normal and reversed directions would have been rotated out of their 180° misalignment.

In the same region of Peru, samples of grey tuffs of Mississippian age were collected and the estimated direction of primary magnetization is given in the table.

Devonian sediments from Uruguay are magnetized in a direction close to the present dipole field whilst red beds dated as Devonian or possibly Silurian (age uncertain) are magnetized in the opposite direction and exhibit "streaking" (see fig. 10).

The most reliable data for the Devonian comes from a sequence of red and purple sandstones from Salta and Jujuy provinces, Argentina. The beds dip in various

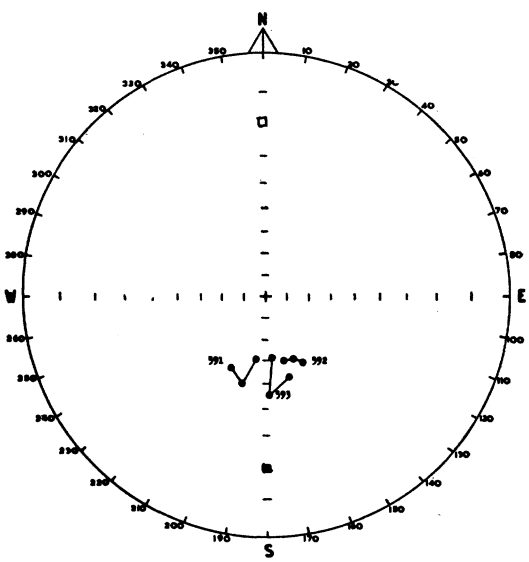


Fig. 9. Directions of n.r.m. of discs from Mississippian red beds from La Rioja province, Argentina.

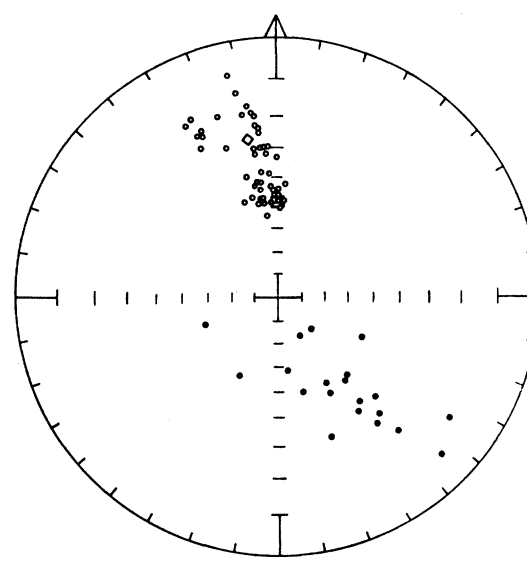


Fig. 10. Directions of n.r.m. of discs of Devonian sediments from Uruguay (the normally magnetized ones) and Parana state, Brazil (the reversely magnetized ones).

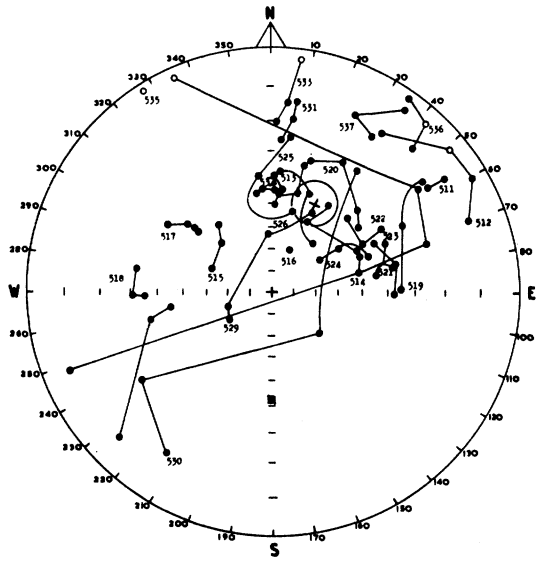
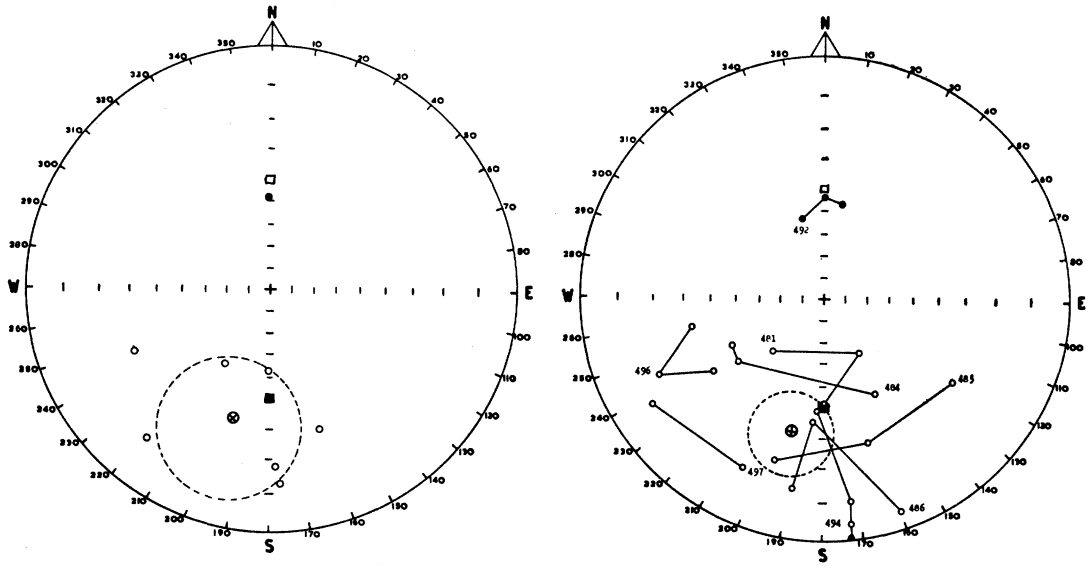


Fig. 11. Directions of n.r.m. of discs of Devonian red beds from Salta and Jujuy provinces, Argentina.

directions, by angles of 30° or 40°. Only one polarity of magnetization occurs but the mean direction makes a large angle with the present field direction (see fig. 11).

A large collection of Cambro-Ordovician and Cambrian red beds from the same area is magnetized in a direction close to the Devonian, but the dip is rather smaller. The scatter of magnetization is greater (see the table). Pre-Cambrian red beds which dip steeply, some of them overturned, were collected from sites in the Andes in Salta province. The directions of magnetization are given in fig. 12, and form one large group



(a) (b)

Fig. 12. Precambrian red beds from Salta province, Argentina.
⊗ mean direction of magnetization;
□ direction of present axial dipole field.
(a) directions of magnetization of hand samples;
(b) directions of magnetization of discs.

ewith the exception of one sample which is magnetized in the opposite direction. The polarity of this sequence of red beds is opposite to that of the Devonian and Cambrian, but the magnetization axis is close to those of the latter formations.

A final result is given for the Precambrian Guabirobá formation from Parana, Brazil.

4. Construction of a Polar Wandering Curve

There appears to have been little relative movement between the present geographic and the palaeomagnetic poles for most of the Mesozoic (see. fig. 13). The youngest formation to give a pole significantly different from the present geographic pole is the Triassic red beds from Argentina, and this movement is confirmed by the results for the Upper Carboniferous or Permian Mitu Formation from Peru. Both of these formations contain normally and reversely magnetized samples and the information obtained from them is considered reliable. At present little data is available for the upper Palaeozoice when a polar movement of about 90° seems to have occurred.

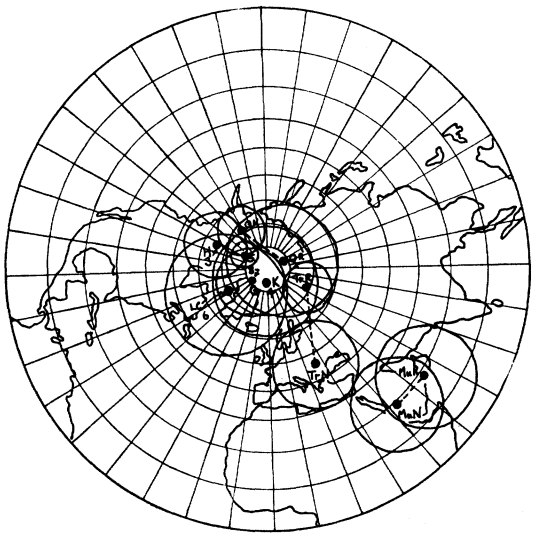


Fig. 13 Pole positions for post Palaeozoic formations from S. America

In fig. 14 a provisional polar wandering curve is taken through

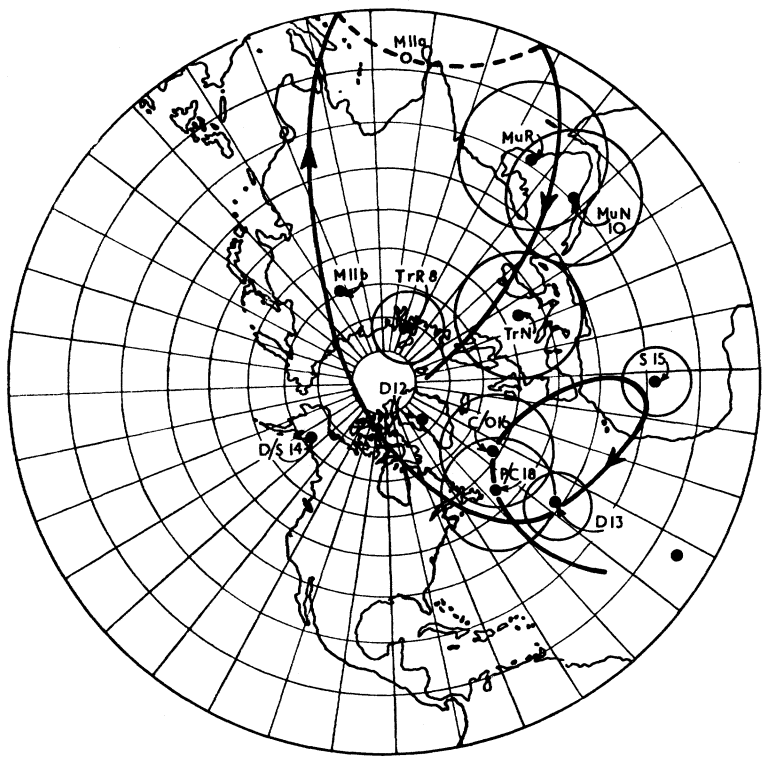


Fig. 14. Provisional curve showing movement of n. pole relative to S. America.

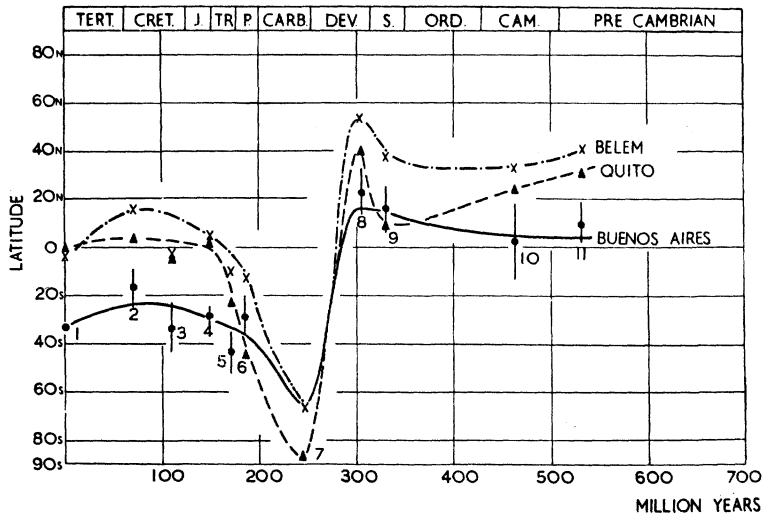


Fig. 15. Palaeolatitudes calculated for Belem, Quito, and Buenos Aires from palaeomagnetic of the following rock formations

1. Present day.
2. Huitrinian purple sandstones Neuquen, Argentina.
3. Kimmeridgian purple sandstones, Neuquen, Argentina.
4. Serra Geral formation, S. Brazil and Uruguay.
5. Triassic red beds, La Rioja, Argentina.
6. Mitu formation, Peru.
7. Mississippian tuffs, Peru.
8. Devonian purple sandstones, Salta and Juyuy, Argentina.
9. Corrego das Pedras formation, Corumba, Brazil.
10. Cambrian purple sandstones, Salta and Juyuy, Argentina.
11. Precambrian purple sandstones, Salta and Juyuy, Argentina.

poles 8 and 10 which are considered reliable and continued through an estimated Mississippian pole no. lla. It is then turned back to pass between Devonian poles 12 and 14 into a group of lower Palaeozoic and Precambrian poles, 13, 15, 16, 18 and 19.

S. America is placed far from the north pole (i.e. very close to the south pole) during the Upper Carboniferous when there is evidence of glaciation, but palaeomagnetic evidence from rocks of this age must be sought to confirm the position of the curve for these times. Unfortunately few of the rocks of this age yet sampled have been sufficiently magnetic.

5. Conclusion.

The reliability of the data presented here will be increased when techniques for reducing scatter of magnetization and eliminating secondary magnetization have been improved. Thermal and alternating field demagnetization work is still being carried out on these and many other rocks from S. America. Additional field work is in progress.

Note added in proof.

A palaeolatitude curve has been constructed for the cities of Belem, Quito and

Buenos Aires from the palaeomagnetic data and is given in fig. 15.

DISCUSSION

T.Y. Ma:

Palaeozoic reef corals from Ordovician to Permian, had flourished widely in arctic regions showing very low latitudinal position.